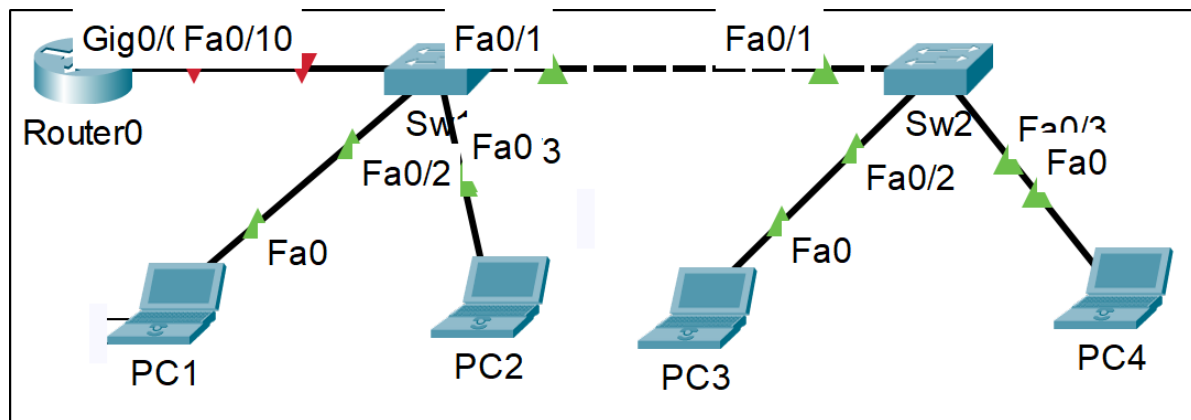


Lab 3 – Inter-VLAN Routing-Home



(This lab is continuation of the earlier lab. Add one router as shown on the diagram and follow the task checklists as below.)

Tasks:

- Configure Fa0/10 interface of Sw1 with static trunk configuration.
- Configure router with two sub interfaces for VLAN 10 and VLAN 20, use the IP details listed below.
 - o Interface Gi0/0.10 – 192.168.1.5/27
 - o Interface Gi0/0.20 – 192.168.1.35/27
- Once configured, all the hosts should be able to reach out to each other.

Detailed Explanation / Configuration

By default, hosts within the same VLAN can communicate with each other since they are part of the same broadcast domain. When a host tries to communicate with the other host, the remote system MAC address should be discovered using ARP broadcast. The ARP broadcast can be spread out within a broadcast domain only. Let's recall that each VLAN is considered as an individual broadcast domain.

We can configure inter-VLAN routing in two ways.

Router-on-a-stick model

In this lab, we are going to check this model where a router is added and connected to any switch available within a switched infrastructure. In order to provide default gateway facility to the hosts, we create different sub-interfaces on the same physical interface and assign one IP address from each VLAN subnet which is configured as a default-gateway on the hosts for the

respective VLANs.

Verification:

```
Sw1#sh inter trunk
```

```
Router#sh ip route connected
```

```
C:\>ping 192.168.1.3
```

```
C:\>ping 192.168.1.42
```

```
C:\>ping 192.168.1.44
```